

Offline, once per re-identification

Excitation data

$$\{x_j, u_j, d_j, x_j^+\}$$

Lift $\Psi(x)$

$$\text{Eq. (12)}$$

EDMD

$$\text{Eq. (14)–(15)}$$

$$A, B_u, B_d, C$$

Koopman linear predictor

$$z^+ = Az + B_u u + B_d d, \quad \hat{y} = Cz$$

$$\text{Eq. (11)}$$

$$\text{measured outputs } y = [S_O, S_{NH}, \text{TN}, S_{N_2O}]^T$$

$$\text{offset-free bias } \hat{b}_k$$

$$\text{Eq. (23)}$$

$$\Phi, \Gamma_u, \Gamma_d$$

Convex quadratic program

$$\min_{U, \varepsilon} \frac{1}{2} \zeta^T P \zeta + q^T \zeta$$

$$\text{s.t. } l \leq A_c \zeta \leq u_c$$

$$\text{Eq. (18), (21); } P > 0$$

energy + fugitive N_2O ,
 NH_4 and TN limits (soft, with back-off)

$$u^* = [K_L a, Q_a]^T$$

one input vector; first block applied
 $K_L a$: oxygen transfer coefficient (d^{-1})
 Q_a : internal recirculation flow ($\text{m}^3 \text{d}^{-1}$)

influent forecast

$$d = [S_{NH}^{\text{in}}, Q]^T$$

measured disturbance — not manipulated

Activated-sludge plant $\dot{x} = f(x, u, d)$
ASM1 kinetics + two-pathway N_2O , Eq. (1)–(6)

Q_a internal recirculation



$K_L a$ applied to the aerated tanks

single-reactor reference model = the one-aerated-tank special case